The role of Camera-Based rPPG in Musical Biofeedback Systems for Stress Reduction and Consciousness Regulation

Fabian Winkelmann

Abstract—Hier kommt der Abstract hin.

Index Terms—rPPG Technique

1 Introduction

Stress can have detrimental effects on the human body, ranging from depression, anxiety and social withdrawal to sleep deprivation and obesity. Stress has a direct impact on the cardiovascular system and therefore influences the heart beat. There are many ways to measure these cardiovascular signals and to track the level of stress. Traditional methods include contact-based sensors, such as chest straps or pulse oximeters. These are very precise but on the other hand, impractical during daily use. To solve this issue, remote Photoplethysmography (rPPG) proposes a solutions to these limitations. This technology uses standard video cameras to monitor the skin color and filter for subtle skilcolor variations that are caused by flowing blood. With the information the heart rate can be calculated [1]. These measurements can be directly reported back to the user using biofeedback. Biofeedback describes a therapeutic concept in which a user receives real-time information about their physiological processes, such as respiration or heart rate. This data is typically made accessible through an output device, such as a visual display. Alternatively, the information can be translated into different forms, such as auditory signals. The primary goal of biofeedback is to make these biological processes perceptible, enabling the user to gain conscious awareness and control over functions that were previously automatic [2].

The objective of this paper is to evaluate such a camera-based stress tracking system combined with musical biofeedback. The goal is to analyze if such a system can be used to lower stress levels and regulate consciousness. In the second chapter theoretical foundations such as stress, consciousness and rPPG techniques are investigated. In the third chapter, case studies will be analyzed. The fourth chapter evaluates whether such a system is theoretically feasible.

2 Theoretical Foundations

This chapter examines the individual components of this reasearch. It starts with a deeper look at stress and its impact on the Autonomic Nervous System, followed by a discussion on heart rate and heart rate variability. Moreover, the concept of rPPG, the mechanism of biofeedback, and the influence of music on stress regulations are investigated.

2.1 Stress and the Autonomic Nervous System

The Autonomic Nervous System (ANS) consists of the parasympathetic and the sympathetic branches. The paraympathikus system act during periods of rest and calm and regulates functions such as "rest and digest". In contrast, the sympathetic system is activated situations in which we precive a threat. In these cases, the heart rate increases to release more energy [3].

McEwen (2007) describes this as the allostatic effect. The body dynamically adjusts its internal state to meet the standards of the environment. For instance, in a life threatening situation a lot of energy is released. The short-term mobilization of energy is a protective reaction of the body. If this activation occurs chronically without enough recovery, it can lead to permenant damage (Figure 1) [4].

Individual differences in stress perception are rooted in the brains' ability to evaluate environmental stimuli. According to McEwen (2007), the brain exhibits significant plasticity, meaning its responses to stressors is shaped by genetics, early development, and past experiences. If an individual has for example experiences a trauma, the brain's system may become hypersensitiv. It might trigger a 'fight or flight' response in harmless situations. This can lead to coping strategies, such as smoking or drinking. These coping stragies try to calm down the parasympathetic branch of the autonomic Nervous System. On the other hand, the plasticity also allows for an intervention in the other direction. The brain can be trained to shift from a threatening state to a calm state [4].

2.2 Heart Beat and Heart Rate Variability

To understand the mechanics of stress detection, it is essential to differentiate between the quantity and the quality of heart activity. These two aspects are represented by Heart Rate (HR) and Heart Rate Variability (HRV) [5].

Heart Rate (HR) is defined as the total number of heartbeats per minute. It acts as a measure of how fast the heart is beating at any given moment. It remains a relatively coarse

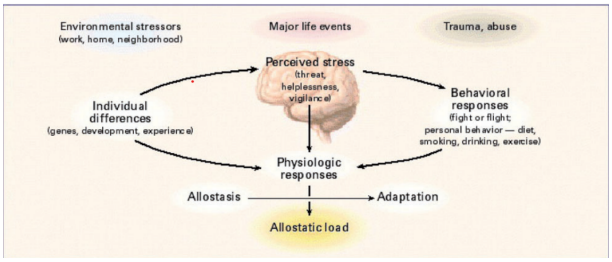

Fig. 1. The Figure illustrates the mechanism of allostatic load and the underlying pathways of stress perception and physiological adaption. [4]

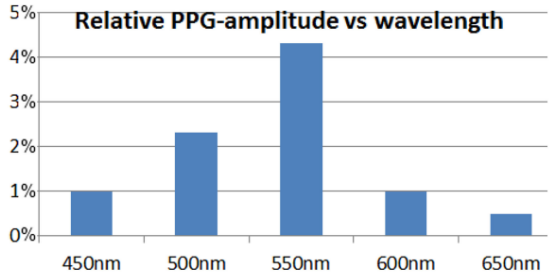

Fig. 2. Spectral sensitivity of the PPG signal. Green light exhibits the highest amplitude, making it the most reliable channel for non-contact heart rate monitoring. [6]

metric. For instance, in an office environment, HR often stays within a specific range, as there is not enough physical activity. This can make it an insensitive marker for subtle cognitive loads or early-stage stress [5].

Heart Rate Variability (HRV), on the other hand, focuses on the smaller level of cardiac timing. It describes the fluctuation in the time intervals between two heartbeats. This is defined as Interbeat Intervals (IBIs). HRV consists of the constant changes in these time intervals from one beat to the next [5].

From a physiological standpoint, a healthy biological system requires to be highly adaptable. This means that a healthy heart does not maintain a perfectly rigid or identical time between intervals. Instead, the autonomic nervous system constantly changes the timing of each beat to respond to internal and external demands [5].

If the intervals between heartbeats become too consistent and lose this natural irregularity, it is often a clinical sign of physiological strain or a lack of regulatory flexibility. High variability indicates that the body is in a state of balance and can rapidly adjust to stressors. Therefore, while HR measures the quantity of beats, HRV reflects the quality and efficiency of the autonomic nervous system's control over the cardiovascular system [5].

2.3 Remote Photoplethysmography (rPPG)

To measure the heart beat and the heart beat variability a technology called remote Photoplethysmography can be employed.

2.3.1 The foundations of rPPG

The foundational work for modern remote photoplethysmography (rPPG) was made by Verkruysse et al. (2008), who demonstrated that the heart beat could be monitored using a standard digital camera. This relies on the fact that hemoglobin absords light differently than the surrounding body tissue. As the blood volume changes in the body with every heartbeat, the intensity of the reflected light varies. Their research identified that the green color channel contains the strongest information regarding the heart beat (Figure 2). Despite the successful proof-of-concept, two challenges were highlighted: Motion artifcats caused by movement and noise due to the camera's sensor [1].

2.3.2 Evolution: From Classic Algorithms to AI-based Detection

Following the initial proof-of-concept, a significant increase in rPPG research was observed, particularly after 2017. This trend is closely linked to the rapid improvements in Artificial Intelligence (AI) and Deep Learning. A recent meta-analysis revealed that in 2025 82.05% [7] of papers about rPPG technology used an approach with artificial intelligence. A major problem is the availability of diverse, large-scale datasets. The available sets only have data of people from North America, Europe and China. This leads to a huge underrepresentation of various age groups and diverse skin tones [7].

2.3.3 Tradtional Frameworks and Region of Interest (ROI)

Traditional rPPG frameworks typically follow a multi-stage process. First, the face is detected and tracked, often using the Viola-Jones algorithm or similar cascade classifiers to define a Region of Interest (ROI). Then methods such as Principal component Analysis (PCA) or Independet Component Analysis (ICA) or employed to isolate the pulse signal from the green color channel. While effective if the person remains still, these approaches struggle with motion artifacts, as they require several seconds of stable data to compute the signal components [7].

While traditional frameworks rely on fixed mathematical models, modern approaches increasingly replace stages of the rPPG pipeline with Deep Learning modules. This is done in the area of face detection and tracking. The Viola-Jones algorithm often struggles with varying lighting conditions and extreme head poses. Current systems frequently utilize more robust AI-based detectors [7].

Furthermore, the region of interest (ROI) significantly impact the accuracy of the heart rate. Although, early studies often used the entire facial area, recent studies recommend using a small rectangular area at the forhead for better results [7].

2.4 Principal Component Analysis

2.4.1 Chrom approach

The Chrominance-based method (CHROM) introduced a physicsbased alternative. It solves the big challenge of subject motion, when calculating the heart beat. Periodic movements can render simple intensity-based selections useless. Furthermore, while many algorithms require long observation intervals to achieve sufficient frequency resolution, CHROM is designed to be highly efficient. Similarly to traditional methods like PCA or ICA, the chrom algorithm also uses the fact that the blood absorb a large percentage

of green light. Additionally it analyzes how the light is reflected on the human skin. According to the dichromatic reflection model this reflection has a Specular and a Diffused Component (Figure 3). Only the diffuse Reflexion enters the skin and has the color of the blood. The specular reflexion does not enter the skin. It is just reflected on the surface and thus does not contain any information about the blood. This specular part is an additive value and thus the division of the two color channels does not cancel this effect out. Thereforce, a subtraktive approach is used [6].

$$(R + noise) - (G + noise) = R - G$$

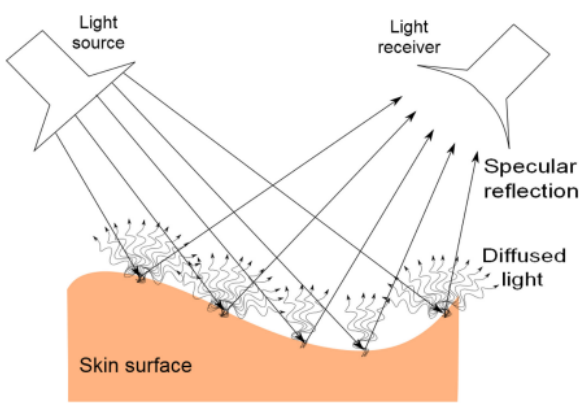

Fig. 3. The illustration depicts light reflection on the skin [6]

The CHROM algorithm projects the normalized RGB signals into two orthogonal chrominance signals, X and Y. In a perfect environment under white light, these signals are mathematically independent. Based on empirical data from a diverse set of subjects, de Haan and Jeanne derived the standardized projection vectors $[0.7682, 0.5121, 0.3841]$, leading to the core transformations:

$$X_s = 3R_n - 2G_n$$

$$Y_s = 1.5R_n + G_n - 1.5B_n$$

To further refine the pulse signal S, these components are combined using the ratio of their standard deviations (α):

$$S = X_s - \alpha \cdot Y_s \quad where \quad \alpha = \frac{\sigma(X_s)}{\sigma(Y_s)}$$

Even after projection, the signal S may contain a slow "drift" caused by respiration or minor posture changes. To isolate the cardiac signal, a Bandpass Filter is applied. The filter is typically tuned to a range of 0.7 Hz to 4.0 Hz, which corresponds to a human heart rate between 42 and 240 BPM. Frequencies outside this window are discarded as non-physiological noise.To extract the final heart rate, a Fast Fourier Transform (FFT) is performed on the filtered signal. The FFT shifts the data from the time domain to the frequency domain, resulting in a power spectrum. The algorithm identifies the "dominant peak" (the highest frequency component) in this spectrum. For instance, a peak at 1.2Hz is converted to the final pulse rate:

$$1.2Hz \times 60seconds = 72BPM$$

One of the most significant advantages of CHROM is its temporal efficiency. While traditional BSS methods (Blind Source Separation) like ICA or PCA often require up to 30 seconds of data to stabilize, CHROM utilizes an Overlap-Add (WOLA) procedure. By using overlapping windows of only 1.6 seconds, the system provides a continuous, near-instantaneous pulse estimate.

In comparative studies, the robustness of CHROM is evident. CHROM has a high stationary accuracy and predictes the correct pulse with a 98% accuracy. It has a 92% correlation with the medical-grade contact sensor. Furthermore, it is resilient against motion. Under havey motion (e.g., on a fitness stepper), maintains a 48% success rate. ICA and PCA fail significantly, and only reach 4% and 11%. Moreover, CHROM can also process different skin tones. Darker Skin Tones naturally lower the Signal-to-Noise Rratio because the absord more light. Chrom remains the most robust classical algorithm across all phenotypes.

2.4.2 Modern Deep Learning Paradigms

With the rapid advancement in artificial intelligence, modern rPPG systems have shifted towards end-to-end models. These architectures can process raw data or image sequences to estimate heart rate directly, often without the need for manual face isolation or region of intrest selection. There are three architectural paradigms: Convolutional Neural Networks (CNNs), Hybrid Neural Networks and Transformed-based architectures [7].

In modern rPPG systems, Convolutional Neural Networks (CNNs) are the standard for processing image data. These networks consist of several layers, mainly an input, an output, and various hidden layers like convolutional and pooling layers. Their main task is to recognize the very subtle color changes in the skin pixels that happen during a cardiac cycle [7].

A big problem in this field, is that there aren't enough large and diverse datasets to train these models from scratch. Because of this, researchers often use Transfer Learning [7]. The idea here is to take a model that was already trained on a different task, such as general face recognition. By 'freezing' the first layers and only fine-tuning the last ones with rPPG-specific data, the model can use its existing knowledge of facial features. This approach makes it possible to get good results even with smaller datasets and saves a lot of training time [8].

In addition, a different category of neural networks has emerged: Hybrid Neural Networks. These combine different neural networks like CNNs, RNNs and autoencoders to form a more robust system [7].

In the past years powerful Large Language Models (LLMS) like ChatGPT have emerged that use transformers to understand words and sentences. The same technique can be employed to analyze pictures. The picture is broken up into small rectangular patches, which the network analyzes in relation to one another [9]. By Comparing different regions simultaneously, the transformer is able to differentiate between the heart beat and noise, such as micro-movements. This leads to a more robust heart rate estimation [7].

2.4.3 Summary of rPPG Frameworks

There are lot of different approaches to calculate the heartbeat with an rPPG technique. Modern Deep Learning Algorithms have emerged in the recent years. They show promising results, although there are not enough diverse datasets. The CHROM Algorithms is promising for a real-time evaluation of data. Traditionally algorithms like PCA are slow and not so precise. Although, there has been a huge advancement in the past years, there are still no long-term studies that monitor the heart beat over several hours. Most researches only test rPPG for a short time, which does not show how it works in real daily life [7].

2.5 Mapping of heart beat to music

RPPG Techniques provide the functionality to measure the heart beat remotely. These measurements then need to be converted to music.

There has been some research on passive music listening without any biofeedback mechanism. According to a meta-analysis by Pelletier(2004), music consistently assists in lowering stress levels. The magnitude of this response is affected by several factors. Patients with prior musical experience, such as playing an instrument, tend to lower their stress levels a lot more. Furthermore, Pelletier notes that indivudal therapy generally results in better relaxation than group session. Although, group sessions are used much more often because they are less expensive. A huge obstacle is that the devices that are used for data collection often cause discomfort. This can lead to false data regarding the relaxation process. Finally, the study suggests that music based on scientific research (e.g., specific rhythmic or frequency patterns) is more effective for stress reduction that music chosen soley based on personal preference [10]. In another study, Linnemann et al. examined how music affects stress in daily life rather than in a laboratory. They came to the conclusion that listening to music can effectively reduce stress levels. Nevertheless, the largest effect is reached when subjects listen to music with then intention of decreasing stress levels. This suggests that the reason why someons listens to music is very important for the actual stress-reducing effect [11].

These results are also supported by another study [12]. They confirm that listening to music alone does not automatically decrease stress levels. Instead, the reason of listening to the music is the most important factor. For example, if someone listens to music to cope with loneliness, it might effectively decrease their feeling of isolation. Nevertheless, if the goal is relaxation, the music only works well if the listener has the specific intention to relax [12].

The study of Bernardi et al. (2006) confirms that the results do not depend on the music preferation of the user rather on the BPM and rhythm structure of the music. Additionally, the study found no evidence of habituation, meaning the users do not simply 'get used' to the music over time. In Contrast to Linnemann et al. (2015), who emphasized the importance of the listener's intention, Bernadi concludes that music can be perceived as a passive form of meditaiton. consequently, this means that the user does not need an intend to reduce stress for measurable results [13].

Passive music listening without any biofeedback system has mixed results. To convert the heartbeat to music, a process called mapping is necessary, that takes the input data and transforms it into music. There are two methods Continous mapping and diskrete mapping. Continous mapping describes the process of constantly changing the music according to the data. During the discrete mapping process certain aspects of the music are changed when a certain threshold is reached [14].

2.5.1 Biofeedback-Driven Regeneration

As explained previously the heart rate is converted to a piece of music. It is now crucial to understand how the music can help the human body to regenerate. According to Lehrer and Gevirtz (2014), the efficiency of biofeedback relies on the synchronization of the cardiovascular and respiration systems [15].

The human cardiovascular system has a certain resonance frequency at which point it exhibts the best effects on the human body. If this state is reached, has maximum internal order and physiological efficiency. This frequency is reached at approximately 0.1 Hz. This is equivalent to roughly six breaths per minute. By breathing at the resonance frequency, heart rate and blood pressure enter a harmonic state. Furthermore, gas exchange in the lungs are also optimised at this state [15].

In addition, the activity of the vagus nerve is stimulated. The vagues nerve is responsible for the communication between the heart and the blood. This makes it possible for the body to lower the heart rate more effectively [15].

Moreover, biofeedback creates a meditative effect that changes the user's state of consciousness. The user has to focus his thoughts and his energy on the biofeedback signal. Thus, it the brain cannot focus on anxious thoughts [15].

2.6 Consciousness and State Regulation

The meditative effect of focusing on one's heart rate or respiration can be used to enhance consciousness. This process is frequently described through the concept of mindfulness. This makes it possible for individuals to develop a sensitivity to their thoughts and emotional states. Bishop et al. (2004) established this definition. A fundamental requirements is described as the 'non-judgemental-approach", meaning. This means that internal states are perceived with curiosity and openness. Internal thoughts and feeling should be accepted as they are [16].

3 CASE STUDIES

In this chapter two case studies will be presented that highlight the effects of using biofeedback. The use different approaches. One uses the heart beat and the other uses EEG. They are both important for validating stress reduction and consciousness regulation.

3.1 The "Unwind" Musical Biofeedback System

The "Unwind" system, developed by Yu et al. (2018), employs a special approach to mapping the heart rate to music. It uses nature soundscapes combined with sedative music. Sedative music is used in music therapy to effectively

decrease heart rate and blood pressure. It has a low tempo and a repetitive structure [17].

The system uses the heart rate variability to provide feedback on the user's breathing patterns. The heart rate accelerates during inhalation and decelerates during exhalation. The framework maps these fluctuations directly to a wind sounscape. The wind volume is reduced during inhalation and during exhalation the wind becomes louder. The user should adapt his breahting pattern in a way that a the wind sound is like a smooth, consistent sinve wave. This promotes consistent respiration [17].

The wind provides immediate feedback on the breathing pattern. The overall relaxation is mapped with a special form of heart rate variability. The system calculates the variability with a sliding window of 16 heartbeats. The organic soundscape is based on the intensity of the Heart Rate Variability. If the variability is low, the environment becomes busy and crowded with bird calls and noises. On the other hand, if the variability increases, the soundscape becomes simpler and more transparent. The frequency of animal sounds decreases and the overall volume drops. This rewards the person with acoustic clarity [17].

The system was validated with 40 young adults. A significant decrease in heart rate was found in the music assisted conditions, and the reduction in respiration rate was significantly greater with biofeedback than without. Skin conductance responses, which indicate sympathetic nervous system activity, decreased markedly during the relaxation sessions. In follow-up interviews, 70% of participants reported that the nature sounds allowed them to quickly visualize calming natural scenes, aiding their focus and mood [17].

3.2 Mindfulness

This study investigated if a system with biofeedback could lower the entry barriers of classical meditation. The focus was on improvements in emotional regulation and cognitive performance. The experiment took place over four-weeks and involved 40 participants that suffered from mild to moderate stress. The participants were separated into an experimental group and an active control group. As a tracking device, this experiment utilized the Muse™ headband (Figure 4). This is a mobile EEG system connected to a smartphone app. The participants had to do daily, active attention traning sessions. The particiapnts were able to gather real-time feedback with the Muse™ headband. The recorded brainwaves were translated into acoustic feedback. The active control group performed identical relaxation exercises for the same duration but without the wearable device or acoustic feedback [18].

The results after four weeks were significant and unserscore the potential of wearables in professional stress management. The control group did not show large changes. On the other hand, the experimental group recorded a decrease in perceived stress of 7% after two weeks. After weeks it went down to 16%. In addition, participants demonstrated higher concentration levels and better impuls control during complex tasks. Moreover, the participants' brainwaves showed a lasting shift toward a relaxed mind. The central finding of Balconi et al. is that real-time feedback is the decisive factor for success.

Fig. 4. Subject wearing the Muse™ EEG headband during a neurofeedback session. [18]

4 Conclusion

Here goes the conclusion

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